

How this book works

This book introduces all the essential concepts needed to understand computer coding. Fun projects throughout put these ideas into practice. Everything is broken down into small chunks so that it's easy to follow and understand.

Each topic is described in detail, with examples and exercises

"See also" boxes list other subjects that are linked to the topic

Pixel people give hints and tips along the way

170 PLAYING WITH PY

BUBBLE BLASTER

Figuring out the distance

In this game, and lots of others, it is often necessary to calculate the distance between two objects. Here's how to do it with a formula to have the computer work it out for you.

11 This function calculates the distance between two objects. Add this bit of code directly to the code you wrote in step 9.

```
from math import sqrt
def distance(id1, id2):
    x1, y1 = get_coords(id1)
    x2, y2 = get_coords(id2)
    return sqrt((x2 - x1)**2 + (y2 - y1)**2)
```

42 STARTING FROM SCRATCH

Hide and seek

Welcome to the special effects studio! Using the purple "Looks" blocks, find out how to make sprites vanish and reappear, grow and shrink, and fade in and out.

Hiding sprites

To make a sprite disappear, use the "hide" block. The sprite is still on the stage, and it can still move around, but it can't be seen unless the "show" block is used to make it visible again.

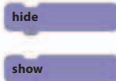
Use the "hide" block to make sprites disappear in games

SEE ALSO

(38-39 Making things move)
Sending (70-71) messages

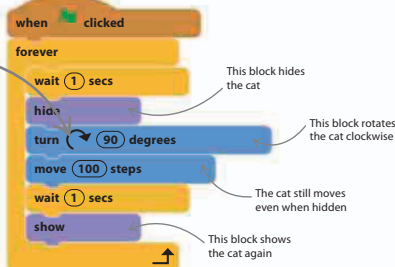
Hide and show

To make a sprite vanish, use the "hide" block. When you're ready for it to be seen again, use the "show" block. These blocks are found in the "Looks" section of the blocks palette.



Disappearing cat

Try this script using the cat sprite. It disappears and reappears but it keeps moving, even when you can't see it.



This block hides the cat

This block rotates the cat clockwise

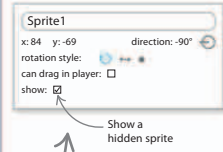
The cat still moves even when hidden

This block shows the cat again

EXPERT TIPS

Showing sprites

Select a sprite in the sprite list. Click the "i" button on it to open the information panel. There you can also use the "show" tick box to show or hide a sprite.



Show a hidden sprite

Sizes and effects

Scripts can be used to change the size of a sprite and add special effects to it.

Choose the type of effect from the dropdown menu. The effect makes the sprite become...

change size by 10

Type in positive numbers to make sprites bigger and negative numbers to make them smaller

set size to 100 %

Higher numbers make sprites bigger and lower numbers make them smaller. 100 is normal size

Changing a sprite's size

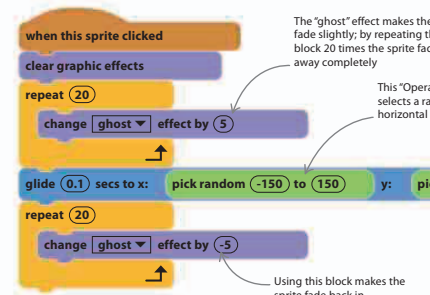
These two blocks can be used to make a sprite bigger or smaller, either by a set amount or by a percentage of its size.

Resets all the effects

Adding effects The graph used to control or distort an experiment

Using effects to teleport

Add a ghost sprite from the "Fantasy" category of the sprite library, and create the script shown below. It makes the ghost appear to teleport when clicked.



The "ghost" effect makes the fade slightly; by repeating the block 20 times the sprite fades away completely

This "Opera" selects a random horizontal position

Using this block makes the sprite fade back in

Instructions show what to click, drag, or select

Labels help explain each step

Colorful illustrations highlight different programming concepts

Programming scripts and code are explained line by line